

Vishal RS

Chennai | vishalrs2325@gmail.com | +91 8838322705 |

linkedin.com/in/vishal-rs | github.com/Vishalrs2325 |

Summary

Highly motivated Electronics and Communication Engineering graduate, currently working as a Product Validation Engineer I at [Cadence Design Systems](#). Skilled in digital logic design, circuit simulation, Verilog, SystemVerilog, and C programming. Proficient in PCB design and industry tools including PSPICE, Xilinx, and KiCad, with a strong interest in hardware validation and system optimization.

Education

• **BE in ECE**, St. Joseph's College of Engineering, Chennai **GPA: 8.50/10** Nov 2021 – May 2025

Skillset

- **Languages:** SystemVerilog, Verilog, C, HTML, CSS
- **Tools and Frameworks:** Visual Studio Code, KiCad 8.0, Xilinx, Pspice, Circuitverse
- **Area of interest:** Digital logics and circuit simulation, PCB design

Certifications

Successfully completed **45 hours** course titled **Comprehensive Electronics and Embedded Systems** provided by IITM Pravartak in **IIT Madras**, Dec 2024.

Successfully completed **12 weeks** online course on **Digital Logics and Circuit simulation** with **92% (Elite Gold)**, conducted by NPTEL.

Research Paper

Title: NEXT HOP SELECTION ALGORITHM USING DS EVIDENCE THEORY FOR WSN

- Our proposed energy efficient algorithm incorporates the DS evidence theory to optimize routing decision.
- Energy efficient and reliable routing algorithm based on weighted DS evidence theory and to find the optimal next hop based on BPA Indexes.

Projects

WATER LEVEL DETECTION, MONITORING AND QUALITY ANALYSIS

- Designed and implemented an automated system for real-time water level detection, monitoring, and quality analysis using an ultrasonic sensor and the system accurately measures water levels in tanks or reservoirs, preventing overflow or shortages and also analyze the acidity of the water using pH sensor.
- **Tools:** Arduino, Embedded C, Ultrasonic sensor, pH sensor, Relay module, Motor

SMART FAN ACTIVATION BASED ON REAL-TIME TEMPERATURE SENSING

- The system monitors ambient temperature and switches the fan on or off accordingly, enhancing energy efficiency and providing a convenient and automated cooling solutions
- **Tools:** Arduino, Embedded C, DHT-22 sensor, Relay module, Motor

Internship

Intern, RAIL INFOTECH SERVICES

2024

- **Braking System:** Learned about the mechanics and operations of locomotive braking systems, including air brakes and dynamic braking for safe and efficient train control.
- **Display System & Pointers:** Knowledge of locomotive display interfaces and pointer systems used to monitor speed, fuel, and operational parameters, ensuring real-time decision-making.
- **Fire Detection Unit:** Understand the functionality of fire detection units in locomotives, capable of identifying early signs of fire hazards, ensuring passenger and crew safety.

Intern, BSNL

2022

- Gained exposure to telecommunications operations and BSNL's role in India's communication network.
- Learned about the evolution of networks from 2G to 4G.